

Wishmitha Samadhi Mendis

Backend / AI Engineer – Python Services, Django APIs and LLM Applications

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SUMMARY

Backend/AI Engineer with an M.Sc. in Artificial Intelligence and Machine Learning (Distinction), about 8 years of professional Python experience and around 3 years building production LLM applications. Strong background in Django/FastAPI backends, APIs and cloud deployment on AWS, alongside applied computer vision research and recent RAG-based AI products.

TECHNICAL SKILLS

Languages: Python, TypeScript, Solidity, Rust, SQL

AI & LLM Engineering: OpenAI API, LlamaIndex, LangGraph, LangChain, Retrieval-Augmented Generation (RAG), Embeddings and Semantic Search, Prompt Engineering

ML & Computer Vision: PyTorch, Keras, Scikit-Learn, OpenCV, NumPy, Pandas

Backend & APIs: Django, FastAPI, Celery, Scrappy

Data & Storage: PostgreSQL, Redis, Memgraph, Pinecone, AWS S3 Buckets

Cloud & DevOps: AWS (ECS, ECR, Amplify), Docker, GitHub Actions CI/CD, Railway, Git, Linux

Spoken Languages: English (Professional), Sinhala (Native)

PROFESSIONAL EXPERIENCE

The PPav

London, United Kingdom

AI Application Software Developer

Aug 2025 – Present, Full-Time

- **TrueWeave:** Built a relationship-intelligence platform for curated communities: it helps organisers understand who their members are and who should meet whom, combining a Django API, a Memgraph knowledge graph, vector search and a Next.js portal.
- Built a LangGraph enrichment pipeline that pulls member details from LinkedIn, checks that the profile belongs to the right person, and flags uncertain matches for human review before writing them back to the backend.
- Built an AI assistant that answers questions about members and events using RAG over the knowledge graph and vector store, and a vector worker that automatically keeps embeddings in sync as member profiles change.
- **Foresight:** Built an NLP/LLM platform that models the full lifecycle of UK R&D tax claims—from gathering client information to drafting the final HMRC report—with a Django backend, document upload and streamed section-by-section generation.
- Designed conversational assistant workflows that guide advisers through complex claim requirements with follow-up questions and context refinement, using semantic retrieval so answers stay aligned with uploaded documents and HMRC guidance.
- Studied and implemented techniques for composing long-form, multi-section reports from incremental LLM outputs, focusing on coherence, consistency and domain adherence, including citation checks before DOCX export.

AxiomFuse

London, United Kingdom

AI/Backend Engineer

Apr 2025 – Aug 2025, Full-Time

- **XCare:** Built a multi-chain digital asset custody and DeFi platform where users can hold, send, swap and earn yield across Ethereum and Solana from one account, using Django REST, PostgreSQL and Redis on AWS ECS.
- Implemented smart accounts on Ethereum (account abstraction) and a custom Solana program in Rust, including guardian-based recovery and fee-taking modules for send, swap and bridge flows.
- Integrated major DeFi protocols on both chains for lending, staking and yield, plus cross-chain bridging and webhook-based deposit detection for Ethereum and Solana transactions.

brdge.ai

London, United Kingdom

AI/Backend Engineer

Feb 2024 – Apr 2025, Full-Time

- Built Generative AI products including LLM assistants and structured document generation platforms, using Retrieval-Augmented Generation with LlamaIndex and LangChain.
- Developed and containerised Django REST APIs for these products, with CI/CD to AWS via ECR, and owned backend and database modelling across client projects.

University of Birmingham

Birmingham, United Kingdom

Postgraduate Teaching Assistant, Computer Science

Jan 2024 – May 2024, Part-Time

- Taught and assessed the Visualisation module on the M.Sc. in Artificial Intelligence and Machine Learning.

Cogniata

Colombo, Sri Lanka

Data Science Consultant

Apr 2022 – Aug 2022, Contract

- **Tea Auction Price Prediction Tool:** Built an automated end-to-end forecasting tool for Sri Lankan tea auction prices, using external factors such as the economic state of producer and importer countries as features via sentiment analysis.
- Collected training data by scraping social media and financial sites, applied NLP sentiment analysis to the raw text, and trained a Long Short-Term Memory (LSTM) time-series model; also worked on related supply-chain disruption forecasting with Apache Airflow automation.

Hong Kong University of Science and Technology

Clear Water Bay, Hong Kong

Graduate Research Assistant

Aug 2019 – Feb 2022, Full-Time

- Researched computer vision and deep learning for 3D reconstruction in the Vision, Graphics and Design group, covering image processing, feature extraction (SIFT), camera geometry, stereo vision, point clouds, Structure from Motion (SfM), Multi-View Stereo (MVS), Neural Radiance Fields (NeRF) and single-view reconstruction.
- **Single-View 3D Foot Reconstruction:** Developed a deep learning network that generates a 3D reconstructed human foot model from a single bottom-view footprint image, using Occupancy Networks.
- **Deep Style Transfer for 3D Objects:** Built the 3D reconstruction module of an end-to-end reconstruction and texturing pipeline where object style can be changed from a reference style image, using Neural Radiance Fields (NeRF). Acknowledged contributor on **StyleCity** (ECCV 2024), a vision-and-text-driven system for styling large-scale 3D urban scene meshes (paper).
- Teaching assistant for Digital Design and Software Engineering undergraduates, and for edX MOOCs on Digital Design, Ergonomics in Design and Ergonomics in Action.

University of Moratuwa

Moratuwa, Sri Lanka

Junior Consultant

Jan 2019 – Aug 2019, Full-Time

- Lectured and ran labs in Machine Learning, Graph Theory, Theory of Computing and Computer Communication, and supervised undergraduate coursework.

WSO2

Colombo, Sri Lanka

Software Engineering Intern

Jul 2017 – Dec 2017, Internship

- **File-based Message Store:** Implemented a file-based NoSQL message store for the WSO2 Message Broker to improve publish/subscribe performance, increasing throughput by about 10% in testing, using Java, OSGi, LevelDB and Redis.

EDUCATION

University of Birmingham

Birmingham, United Kingdom

M.Sc. Artificial Intelligence and Machine Learning – Distinction (80.55%)

Sep 2022 – Sep 2023

- **Key modules:** Mathematical Foundations of AI and ML (94%), Machine Learning and Data Analysis (90%).
- **Thesis – Detecting Ink from Ancient Documents:** Used deep learning and computer vision to extract hidden ink signals from carbonised Herculaneum papyri scrolls, so scholars can read the text without destroying the scrolls. Employed U-Net image segmentation and transfer learning (PyTorch) (GitHub).

University of Moratuwa

Moratuwa, Sri Lanka

B.Sc. Eng. (Hons.) Computer Science and Engineering – First Class (GPA 3.81/4.20)

Nov 2014 – Dec 2018

PUBLICATIONS

Educational Data Mining: A Review on Data Collection Process | *International Conference on Advances in ICT for Emerging Regions (ICTer)*, Sep 2018

Proposed a framework with common guidelines for effective data collection in educational data mining research, from the undergraduate thesis “*Data-Driven Instruction Strategies for Sri Lankan Schools*”.